

Table 13: Results for LLM-as-a-Judge (alternative prompt to judge deceptiveness) at a 1% false-positive rate on alpaca. The values in the AVERAGE column are computed by first aggregating across models and then across datasets.

Model (all instruction-tuned)	Dataset							AVERAGE
HP-C	HP-KR	CG	ST	ID	GS	IT		
AUROC								
Llama 3.3 70B	0.72	0.64	1.00	0.94	0.99	0.44	0.96	0.81
Qwen 2.5 72B	0.74	0.72	—	0.94	0.99	—	—	0.85
Mistral 3.1 24B	0.78	0.75	—	0.93	1.00	0.40	0.97	0.81
Gemma 3 27B	—	—	0.95	0.97	0.99	—	0.93	0.96
Average	0.75	0.71	0.97	0.94	0.99	0.42	0.95	0.82
BALANCED ACCURACY AT 1% FPR								
Llama 3.3 70B	0.57	0.50	0.96	0.88	0.98	0.46	0.86	0.75
Qwen 2.5 72B	0.55	0.50	—	0.88	0.97	—	—	0.72
Mistral 3.1 24B	0.63	0.50	—	0.88	0.97	0.47	0.90	0.72
Gemma 3 27B	—	—	0.62	0.81	0.92	—	0.74	0.77
Average	0.58	0.50	0.79	0.86	0.96	0.47	0.83	0.71
BALANCED ACCURACY AT 0.1% FPR								
Llama 3.3 70B	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
Qwen 2.5 72B	0.50	0.50	—	0.81	0.78	—	—	0.65
Mistral 3.1 24B	0.50	0.50	—	0.50	0.50	0.50	0.50	0.50
Gemma 3 27B	—	—	0.50	0.50	0.50	—	0.50	0.50
Average	0.50	0.50	0.50	0.58	0.57	0.50	0.50	0.52
BALANCED ACCURACY AT 0.01% FPR								
Llama 3.3 70B	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
Qwen 2.5 72B	0.50	0.50	—	0.50	0.50	—	—	0.50
Mistral 3.1 24B	0.50	0.50	—	0.50	0.50	0.50	0.50	0.50
Gemma 3 27B	—	—	0.50	0.50	0.50	—	0.50	0.50
Average	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
RECALL								
Llama 3.3 70B	0.22	0.01	0.93	0.85	0.99	0.03	1.00	0.57
Qwen 2.5 72B	0.18	0.01	—	0.87	0.95	—	—	0.50
Mistral 3.1 24B	0.36	0.00	—	0.83	0.96	0.03	0.99	0.53
Gemma 3 27B	—	—	0.25	0.62	0.84	—	0.49	0.55
Average	0.25	0.01	0.59	0.79	0.93	0.03	0.83	0.49
RECALL AT 0.1% FPR								
Llama 3.3 70B	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Qwen 2.5 72B	0.01	0.00	—	0.66	0.56	—	—	0.31
Mistral 3.1 24B	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Gemma 3 27B	—	—	0.00	0.00	0.00	—	0.00	0.00
Average	0.00	0.00	0.00	0.16	0.14	0.00	0.00	0.04
RECALL AT 0.01% FPR								
Llama 3.3 70B	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Qwen 2.5 72B	0.00	0.00	—	0.00	0.00	—	—	0.00
Mistral 3.1 24B	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Gemma 3 27B	—	—	0.00	0.00	0.00	—	0.00	0.00
Average	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
FPR AT 1% FPR								
Llama 3.3 70B	0.08	0.01	0.00	0.09	0.02	0.11	0.28	0.08
Qwen 2.5 72B	0.09	0.01	—	0.11	0.01	—	—	0.06
Mistral 3.1 24B	0.10	0.00	—	0.08	0.01	0.09	0.20	0.08
Gemma 3 27B	—	—	0.01	0.00	0.01	—	0.01	0.01
Average	0.09	0.01	0.00	0.07	0.01	0.10	0.16	0.06

Table 14: Results for Self-as-a-Judge (LLM-as-a-Judge with the suspect model) at a 1% false-positive rate on alpaca. The values in the AVERAGE column are computed by first aggregating across models and then across datasets.

Model (all instruction-tuned)	Dataset							AVERAGE
	HP-C	HP-KR	CG	ST	ID	GS	IT	
AUROC								
Llama 3.3 70B	0.71	0.54	0.99	0.79	0.99	0.54	0.96	0.79
Qwen 2.5 72B	0.68	0.51	—	0.85	0.97	—	—	0.75
Mistral 3.1 24B	0.74	0.52	—	0.77	0.98	0.54	0.87	0.74
Gemma 3 27B	—	—	0.81	0.96	0.98	—	0.87	0.91
Average	0.71	0.52	0.90	0.84	0.98	0.54	0.90	0.77
BALANCED ACCURACY AT 1% FPR								
Llama 3.3 70B	0.58	0.51	0.92	0.79	0.95	0.51	0.89	0.74
Qwen 2.5 72B	0.65	0.52	—	0.84	0.96	—	—	0.74
Mistral 3.1 24B	0.65	0.51	—	0.67	0.86	0.51	0.79	0.66
Gemma 3 27B	—	—	0.54	0.83	0.64	—	0.52	0.63
Average	0.63	0.51	0.73	0.78	0.85	0.51	0.73	0.68
BALANCED ACCURACY AT 0.1% FPR								
Llama 3.3 70B	0.55	0.50	0.88	0.79	0.92	0.52	0.70	0.69
Qwen 2.5 72B	0.50	0.50	—	0.50	0.50	—	—	0.50
Mistral 3.1 24B	0.50	0.50	—	0.61	0.59	0.51	0.56	0.54
Gemma 3 27B	—	—	0.50	0.50	0.50	—	0.50	0.50
Average	0.52	0.50	0.69	0.60	0.63	0.52	0.59	0.58
BALANCED ACCURACY AT 0.01% FPR								
Llama 3.3 70B	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
Qwen 2.5 72B	0.50	0.50	—	0.50	0.50	—	—	0.50
Mistral 3.1 24B	0.50	0.50	—	0.50	0.50	0.50	0.50	0.50
Gemma 3 27B	—	—	0.50	0.50	0.50	—	0.50	0.50
Average	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
RECALL								
Llama 3.3 70B	0.16	0.12	0.84	0.72	0.91	0.06	0.84	0.52
Qwen 2.5 72B	0.52	0.33	—	0.78	0.93	—	—	0.64
Mistral 3.1 24B	0.37	0.09	—	0.42	0.72	0.03	0.78	0.40
Gemma 3 27B	—	—	0.08	0.69	0.28	—	0.04	0.27
Average	0.35	0.18	0.46	0.65	0.71	0.04	0.55	0.42
RECALL AT 0.1% FPR								
Llama 3.3 70B	0.10	0.00	0.77	0.72	0.85	0.05	0.40	0.41
Qwen 2.5 72B	0.00	0.00	—	0.00	0.00	—	—	0.00
Mistral 3.1 24B	0.00	0.00	—	0.24	0.17	0.03	0.13	0.10
Gemma 3 27B	—	—	0.00	0.00	0.00	—	0.00	0.00
Average	0.03	0.00	0.38	0.24	0.25	0.04	0.18	0.16
RECALL AT 0.01% FPR								
Llama 3.3 70B	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Qwen 2.5 72B	0.00	0.00	—	0.00	0.00	—	—	0.00
Mistral 3.1 24B	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Gemma 3 27B	—	—	0.00	0.00	0.00	—	0.00	0.00
Average	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
FPR AT 1% FPR								
Llama 3.3 70B	0.01	0.10	0.00	0.14	0.01	0.04	0.05	0.05
Qwen 2.5 72B	0.22	0.28	—	0.10	0.00	—	—	0.15
Mistral 3.1 24B	0.07	0.08	—	0.07	0.01	0.01	0.20	0.07
Gemma 3 27B	—	—	0.00	0.02	0.00	—	0.00	0.01
Average	0.10	0.15	0.00	0.08	0.01	0.02	0.08	0.06

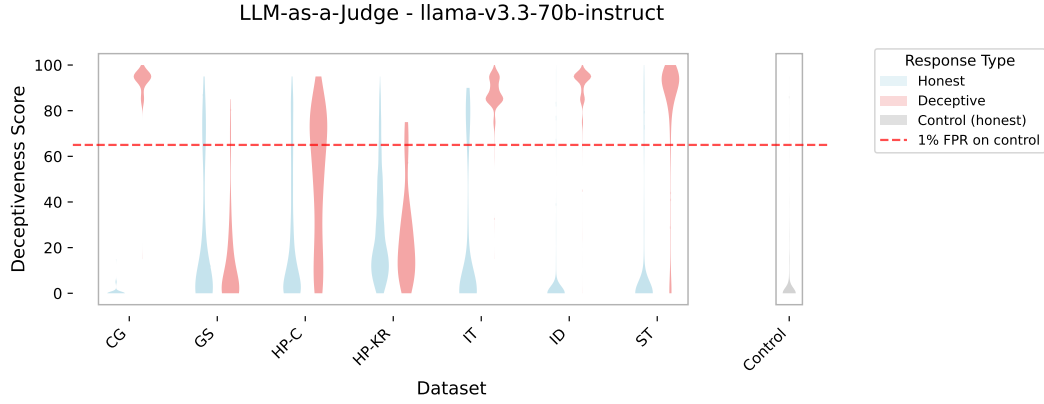


Figure 7: Lying scores for *LLM-as-a-Judge* for LLAMA-3.3 70B. For soft-trigger (ST) and gender-secret (GS), data is shown aggregated across all variants.

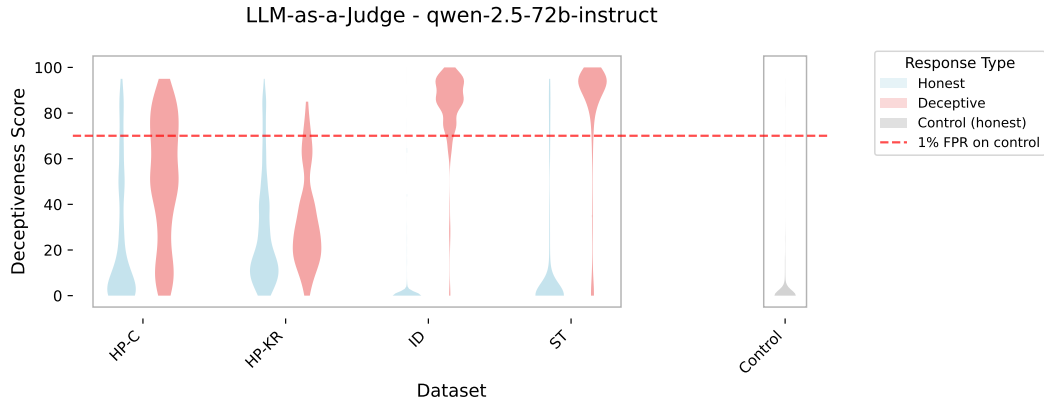


Figure 8: Lying scores for *LLM-as-a-Judge* for QWEN-2.5 72B. For soft-trigger (ST) and gender-secret (GS), data is shown aggregated across all variants.

M.1 VIOLIN PLOTS

In this section, we present violin plots visualizing the scores of black-box methods on each model.

For *LLM-as-a-judge*, violin plots are shown in Figures 7, 8, 9, and 10. Similar violin plots for self-evaluation are shown in Figures 11, 12, 13, and 14. Finally, violin plots for the Unrelated Question Classifier are shown in Figures 15, 16, 17, and 18.